MA0367 TIME SERIES ANALYSIS AND FORECASTING
HOMEWORK 6

Q1) Consider the causal AR(2) process $x_t = x_{t-1} - 0.5x_{t-2} + \varepsilon_t$, where ε_t is white noise, $t \in \mathbb{Z}$. Find the coefficients c_0, c_1, c_2, c_3, c_4 of the MA representation $x_t = \sum_{j=0}^{\infty} c_j \varepsilon_{t-j}$ of this AR(2) process.

We have the causal AR(2) process
$x_t - x_{t-1} + 0.5x_{t-2} = \varepsilon_t$.

Since the AR equation can be written as $(1 - B + 0.5B^2)X_t = \varepsilon_t$ and $\frac{1}{1-u} = 1 + u + u^2 + u^3 + \dots$ (by the taylor expansion), we have

$$X_t = \frac{1}{1 - (B - 0.5B^2)} \varepsilon_t$$

$$= (1 + (B - 0.5B^2) + (B - 0.5B^2)^2 + (B - 0.5B^2)^3 + (B - 0.5B^2)^4 + \dots)\varepsilon_t$$

Since we are interested in terms with B^j, $j \le 4$, we have

$$X_t = 1 + B - 0.5B^2 + B^2 - 2 \cdot 0.5BB^2 + 0.5^2B^4 + B^3 - 0.5B^4 - B^4 - 0.5B^5 + 0.25B^5 - 0.125B^6 + B^4 + \dots$$

$$= 1 + B + 0.5B^2 - 0.25B^4 + \dots$$

$$= \varepsilon_t + \varepsilon_{t-1} + 0.5\varepsilon_{t-2} + 0.\varepsilon_{t-3} - 0.25\varepsilon_{t-4} + \dots$$

Thus, $c_0 = 1$, $c_1 = 1$, $c_2 = 0.5$, $c_3 = 0$, $c_4 = -0.25$

Q2) Consider the causal AR(2) process $X_t = X_{t-1} - 0.4X_{t-2} + 0.3X_{t-3} + \varepsilon_t$ where ε_t is white noise, $t \in \mathbb{Z}$. Find the coefficients c_0, c_1, c_2, c_3, c_4 of the MA representation $X_t = \sum_{j=0}^{\infty} c_j \varepsilon_{t-j}$ of this AR(3) process.

We have the causal AR(3) process

$X_t - X_{t-1} + 0.4X_{t-2} - 0.3X_{t-3} + \varepsilon_t$.

Since the AR equation can be written as $(1 - B + 0.4B^2 - 0.3B^3)X_t = \varepsilon_t$ and $\frac{1}{1-u} = 1 + u + u^2 + u^3 + u^4 + \dots$ (by the taylor expansion, we have

$$X_t = \frac{1}{1 - (B - 0.4B^2 + 0.3B^3)} \varepsilon_t$$

$$= (1 + (B - 0.4B^2 + 0.3B^3) + (B - 0.4B^2 + 0.3B^3)^2 + (B - 0.4B^2 + 0.3B^3)^3 + (B - 0.4B^2 + 0.3B^3)^4 + \dots)\varepsilon_t$$

Since we are interested in terms with B^j, $j \leq 4$, we have

$$X_t = (1 + B - 0.4B^2 + 0.3B^3 + B^2 - 0.8B^3 + 0.6B^4 + 0.16B^4 - 0.24B^5 + 0.09B^6 + 0.027B^9 - 0.108B^8 + 0.414B^7 - 0.784B^6 + 1.38B^5 - 1.2B^4 + B^3 + B^4 + \dots)\varepsilon_t$$

$$= (1 + B + 0.6B^2 + 0.5B^3 + 0.56B^4 + \dots)\varepsilon_t$$

$$= \varepsilon_t + \varepsilon_{t-1} + 0.6\varepsilon_{t-2} + 0.5\varepsilon_{t-3} + 0.56\varepsilon_{t-4} + \dots$$

Thus $c_0 = 1$, $c_1 = 1$, $c_2 = 0.6$, $c_3 = 0.5$, $c_4 = 0.56$

Q3) Find the covariance function for a discrete weakly stationary process X_t with spectral density $f(\lambda) = 2 + \lambda^2$.

The covariance function is computed by

$$R(k) = \int_{-\pi}^{\pi} \cos(k\lambda) f(\lambda)\, d\lambda$$

Thus,

$$R(0) = \int_{-\pi}^{\pi} (2 + \lambda^2)\, d\lambda = \left[2\lambda + \frac{\lambda^3}{3}\right]_{-\pi}^{\pi}$$

$$= \left(2\pi + \frac{\pi^3}{3}\right) - \left(-2\pi - \frac{\pi^3}{3}\right)$$

$$= 4\pi + \frac{2\pi^3}{3}$$

For $k > 0$, we have

$$R(k) = \int_{-\pi}^{\pi} 2\cos(k\lambda)\, d\lambda + \int_{-\pi}^{\pi} \cos(k\lambda)\lambda^2\, d\lambda$$

$$= 0 + 2\int_{0}^{\pi} \cos(k\lambda)\lambda^2\, d\lambda \quad \text{(due to symmetry)}$$

$$= +2\int_{-\pi}^{\pi} \cos(k\lambda)\lambda^2\, d\lambda$$

let $u = \lambda^2$ $\quad \frac{dv}{d\lambda} = \cos(k\lambda)$

$\frac{du}{d\lambda} = 2\lambda$ $\quad v = \frac{1}{k}\sin(k\lambda)$

$$= +2\left[\frac{\lambda^2}{k}\sin(k\lambda) - \frac{2}{k}\int \lambda \sin(k\lambda)\, d\lambda\right]_0^{\pi}$$

let $u = \lambda$ $\quad \frac{dv}{d\lambda} = \sin(k\lambda)$

$\frac{du}{dv} = 1$ $\quad v = -\frac{1}{k}\cos(k\lambda)$

$$= +2\left[\frac{\lambda^2}{k}\sin(k\lambda) - \frac{2}{k}\left(-\frac{\lambda}{k}\cos(k\lambda) + \frac{1}{k}\int \cos(k\lambda)\, d\lambda\right)\right]_0^{\pi}$$

$$= +2\left[\frac{\lambda^2}{k}\sin(k\lambda) + \frac{2\lambda}{k^2}\cos(k\lambda) - \frac{2}{k^3}\sin(k\lambda)\right]_0^{\pi}$$

$$= +2\left[\left[\frac{\pi^2}{k}\sin(\pi k) + \frac{2\pi}{k^2}\cos(\pi k) - \frac{2}{k^3}\sin(\pi k)\right] - \left[0 + 0 - \frac{2}{k^3}\sin(0)\right]\right]$$

$$= +\frac{4\pi}{k^2}\cos(\pi k) = +\frac{(-1)^k 4\pi}{k^2}$$